

Electrical fundamentals

1. Element Constitutive Equations

Ohm's Law

$$v = iR$$

Capacitor

$$i = C \frac{dv}{dt}$$

Inductor

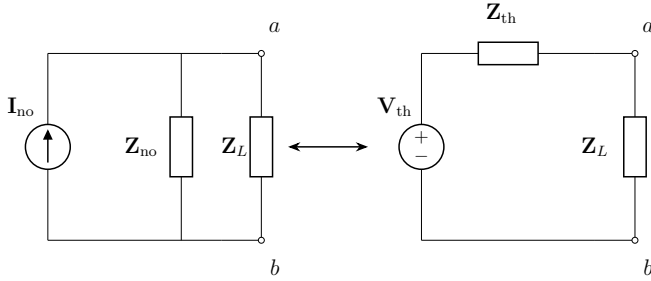
$$v = L \frac{di}{dt}$$

Shockley's Equation

$$i = I_s \left(e^{v/nV_T} - 1 \right)$$

2. Source Transformation

Thevenin-Norton Theorem



$$I_{no} = \frac{V_{th}}{Z_{th}}, \quad Z_{th} = Z_{no}$$

3. Electrostatics

Coulomb's Law

$$\mathbf{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|}$$

Point Charge Q at O

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \mathbf{a}_r$$

Infinite line (ρ_l) on z-axis

$$\mathbf{E} = \frac{\rho_l}{2\pi\epsilon_0 r} \mathbf{a}_\rho$$

Finite line (ρ_l) on z-axis

$$\mathbf{E} = \frac{\rho_l}{4\pi\epsilon_0} (\sin \theta_1 + \sin \theta_2) \mathbf{a}_\rho$$

Infinite sheet (ρ_s) on $z = 0$

$$\mathbf{E} = \pm \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z$$

Ring (Q), radius ρ on $z = 0$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{zQ}{(\rho^2 + z^2)^{3/2}} \mathbf{a}_z$$

Cylinder (ρ_v), radius a on z-axis

$$\mathbf{E} = \begin{cases} \frac{\rho_v \rho}{2\epsilon_0} \mathbf{a}_\rho, & \rho < a \\ \frac{\rho_v a^2}{2\epsilon_0 \rho} \mathbf{a}_\rho, & \rho > a \end{cases}$$

Sphere (ρ_v) with radius a

$$\mathbf{E} = \begin{cases} \frac{\rho_v r}{3\epsilon_0} \mathbf{a}_r, & r < a \\ \frac{\rho_v a^3}{3\epsilon_0 r^2} \mathbf{a}_r, & r > a \end{cases}$$

4. Magnetostatics

Biot-Savart Law

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \hat{\mathbf{r}}}{r^2}$$

Infinite filament $\rho = 0$, dir. $\mathbf{a}_z$

$$\mathbf{B} = \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_\phi$$

Finite filament $\rho = 0$, dir. $\mathbf{a}_z$

$$\mathbf{B} = \frac{\mu_0 I}{4\pi\rho} (\sin \alpha_1 - \sin \alpha_2) \mathbf{a}_\phi$$

Sheet, line density $\mathbf{K}$, norm. $\hat{\mathbf{n}}$

$$\mathbf{B} = \frac{1}{2} \mu_0 \mathbf{K} \times \hat{\mathbf{n}}$$

Circular loop on $z = 0$, dir. $\mathbf{a}_z$

$$\mathbf{B} = \frac{\mu_0 I}{2} \frac{\rho^2}{(z^2 + \rho^2)^{3/2}} \mathbf{a}_z$$

Cable with radius a , dir. $\mathbf{a}_z$

$$\mathbf{B} = \begin{cases} \frac{\mu_0 I \rho}{2\pi a^2} \mathbf{a}_\phi, & \rho < a \\ \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_\phi, & a < \rho < b \end{cases}$$

Long solenoid, radius a on $z = 0$

$$\mathbf{B} = \begin{cases} \mu_0 n I \mathbf{a}_z, & \rho < a \\ \mathbf{0}, & \rho > a \end{cases}$$

5. Maxwell's Equations

Lorentz Force

$$\mathbf{F}_L = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Gauss's Law for E-field

$$\oiint_S \mathbf{E} \cdot d\mathbf{A} = \frac{q}{\epsilon_0}$$

Gauss's Law for B-field

$$\oiint_S \mathbf{B} \cdot d\mathbf{A} = 0$$

Faraday's Law

$$\oint_C \mathbf{E} \cdot d\mathbf{L} = - \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{A}$$

Ampère-Maxwell Law

$$\oint_C \mathbf{B} \cdot d\mathbf{L} = \mu_0 I + \mu_0 \epsilon_0 \iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{A}$$